

A Formulation

conventional workflow

scientific choices supplied upstream

coordinates + library

PySINDy relation fit

SIR

scientific object $O = (D, \Omega_{obs}, S, E, \Pi, A)$

discovery contract $K_q = (q, I_q, C_q, R_q, U_q, V_q)$

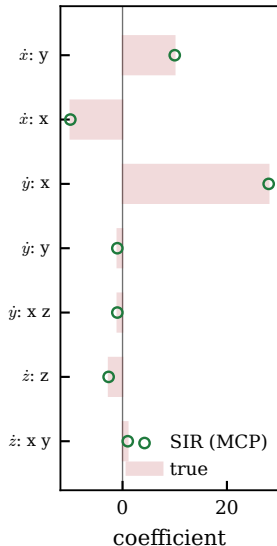
search over qualified (C, R)

PySINDy as one solver $R_q(C)$

qualified representation C_q^*

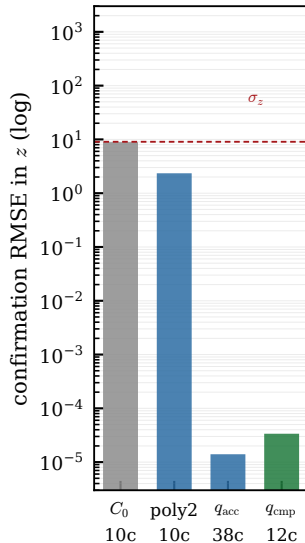
*the estimator is nested inside the contract,
not opposed to it*

B Containment



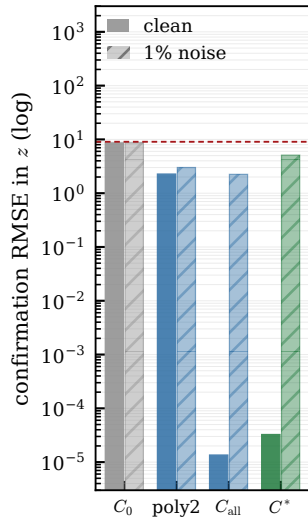
exact support, all three equations
 $\max |\Delta c| \text{ vs truth } \approx 2 \times 10^{-3}$

C Two contracts



c = coordinate count
 accuracy and compactness
 disagree here

D Uncertainty



q_{rob} = same coordinates as q_{acc} ;
 uncertainty changed the
 qualification, not the coordinates